

XIANYANG, China

WMO: 570360

Lat: 34.447N

Lon: 108.752E

Elev: 1572

StdP: 13.88

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 16.0	(c) 19.5	(d) -0.8	(e) 5.6	(f) 32.6	(g) 3.2	(h) 6.9	(i) 32.1	(j) 16.7	(k) 36.1	(l) 14.8	(m) 34.0	(n) 2.7	(o) 270	(p) 0.435

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.0	98.2	73.9	95.2	73.5	92.9	73.0	79.6	90.6	78.2	88.6	76.9	86.8	6.7	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.0	149.1	85.6	75.3	140.7	84.5	73.7	133.1	82.9	44.8	91.5	43.2	88.7	41.6	87.1	85.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 18.4	(b) 15.8	(c) 13.6	(e) 11.2	(f) 102.8	(g) 5.4	(h) 2.0	(i) 7.3	(j) 104.2	(k) 4.1	(l) 105.4	(m) 1.1	(n) 106.5	(o) -2.8	(p) 108.0		
(a) 18.4	(b) 15.8	(c) 13.6	(e) 9.7	(f) 82.1	(g) 4.9	(h) 1.7	(i) 6.1	(j) 83.3	(k) 3.3	(l) 84.3	(m) 0.5	(n) 85.3	(o) -3.0	(p) 86.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	58.0	31.8	38.9	49.5	60.3	69.1	77.8	81.5	78.4	69.1	58.1	45.4	34.6		
	DBStd	17.99	5.62	6.42	7.15	6.94	5.85	5.62	5.32	5.68	6.12	5.90	6.38	5.14		
	HDD50	1630	565	312	97	7	0	0	0	0	7	165	477			
	HDD65	4226	1030	730	481	173	28	2	0	1	25	225	589	942		
	CDD50	4537	0	2	82	315	591	835	976	880	573	257	26	0		
	CDD65	1656	0	0	0	30	154	387	511	415	148	11	0	0		
	CDH74	17561	0	0	34	410	1568	4412	5887	4066	1085	99	0	0		
	CDH80	8048	0	0	4	105	574	2232	2982	1780	356	15	0	0		
	WSAvg	5.3	4.3	5.2	6.1	6.0	5.8	5.8	6.3	6.1	5.4	4.4	4.2	4.1		
	PrecAvg	20.8	0.2	0.4	0.9	1.4	2.2	2.4	3.4	3.3	3.5	2.0	0.9	0.2		
(15) Precipitation	PrecMax	34.1	0.7	1.1	2.3	4.0	5.6	4.4	8.2	7.7	11.8	5.0	2.6	0.7		
	PrecMin	11.6	0.0	0.0	0.0	0.2	0.1	0.2	0.2	0.4	0.6	0.4	0.0	0.0		
	PrecStd	4.8	0.2	0.3	0.6	0.9	1.4	1.2	1.7	1.8	2.5	1.1	0.7	0.2		
	PrecStd	4.8	0.2	0.3	0.6	0.9	1.4	1.2	1.7	1.8	2.5	1.1	0.7	0.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.2	62.8	78.1	88.0	94.6	100.7	102.2	98.6	93.6	81.6	69.5	57.1		
		MCWB	40.6	45.7	57.4	64.9	65.8	69.8	76.3	77.5	73.8	62.3	52.8	42.3		
	2%	DB	49.5	58.6	73.0	82.6	89.5	97.2	98.5	95.0	88.0	76.7	64.4	51.8		
		MCWB	37.3	44.1	54.5	62.7	65.3	70.0	75.8	76.8	71.2	60.7	51.6	39.5		
	5%	DB	45.0	54.0	68.0	78.8	86.1	94.8	95.4	91.7	84.0	73.0	60.6	48.2		
		MCWB	34.9	42.1	51.8	61.3	64.9	69.5	75.7	76.0	69.7	59.9	49.6	37.9		
	10%	DB	42.2	50.4	63.0	74.8	82.7	91.5	93.0	89.2	80.4	69.2	57.0	44.8		
		MCWB	33.5	40.1	49.5	59.2	64.4	68.8	75.0	75.3	67.9	58.4	48.3	36.4		
	0.4%	WB	41.1	48.5	60.3	69.7	71.8	77.3	81.9	81.6	77.4	66.7	56.3	44.0		
		MCDB	53.1	59.0	72.5	81.8	84.2	89.4	93.3	91.7	88.2	74.2	63.8	51.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	38.2	45.6	56.0	65.7	69.7	75.0	80.0	79.6	74.3	64.3	53.7	41.2		
		MCDB	46.7	55.0	69.1	77.9	82.0	87.8	91.6	89.8	83.5	70.9	59.8	48.9		
	5%	WB	36.0	43.5	53.4	63.0	67.9	73.4	78.6	78.2	71.6	62.3	51.8	39.1		
		MCDB	43.7	51.5	64.7	75.0	80.2	86.0	89.9	87.7	79.7	69.1	58.7	45.7		
	10%	WB	34.1	41.5	50.8	60.7	66.3	71.7	77.3	76.9	69.7	60.3	49.6	37.2		
		MCDB	40.8	49.1	61.6	71.5	78.1	84.9	87.9	85.8	77.0	66.6	55.4	43.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	17.3	18.3	20.5	21.7	21.8	21.5	18.0	16.4	16.0	17.1	17.7	17.7		
		MCDBR	23.8	25.7	29.7	29.8	29.2	28.7	23.2	20.7	23.0	24.8	24.9	24.8		
		MCWBR	14.6	14.9	15.5	14.1	11.8	9.4	7.9	8.0	9.7	11.6	13.8	14.9		
		MCDBR	21.1	22.5	26.8	25.8	23.2	20.7	18.5	17.6	17.9	19.2	20.9	21.6		
	5% WB	MCWBR	13.4	13.7	14.6	13.7	11.0	8.6	7.9	8.1	9.1	10.0	12.3	13.7		
		MCWBR	13.4	13.7	14.6	13.7	11.0	8.6	7.9	8.1	9.1	10.0	12.3	13.7		
(40) Clear-Sky Solar Irradiance	taub		0.541	0.621	0.637	0.629	0.612	0.638	0.661	0.698	0.652	0.630	0.573	0.509		
	taud		1.699	1.572	1.574	1.624	1.721	1.711	1.708	1.620	1.674	1.658	1.700	1.779		
	Ebn at Noon		194	194	207	220	226	220	213	202	202	189	184	194		
	Edh at Noon		60	76	82	81	75	75	75	81	73	69	60	52		
	RadAvg		801	942	1287	1533	1642	1705	1630	1416	1104	953	814	752		
(45) All-Sky Solar Radiation	RadStd		88	152	126	150	129	157	209	200	125	147	117	101		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		-0.72	-4.07	-2.50	N/A	N/A	N/A	+110	+142	N/A	-128	
(47) Regional (7 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page